Derivatives: Power, Product, and Quotient Rules

Power rule with rewritten radicals and negative exponents, product rule, quotient rule, and one derivative from the limit definition

Directions.

- Name the rule you are using before you use it, then show the substitution into that rule.
- Rewrite radicals and reciprocals as powers ($\sqrt{x} = x^{1/2}$, $\frac{1}{x^n} = x^{-n}$) *before* differentiating.
- Simplify each answer, but do not expand a squared denominator.
- Product rule: $(uv)' = u'v + uv'$. Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ — the numerator order matters.

1. Find $f'(x)$. $f(x) = 4x^5 - 3x^2 + 7$

Answer: ____________________

2. Find $g'(x)$. Rewrite each term as a power of x first. $g(x) = 6\sqrt{x} + \dfrac{5}{x^2}$

Answer: ____________________

3. Use the **limit definition** of the derivative to find $f'(x)$ for $f(x) = x^3$. Expand the numerator, cancel the h, then take the limit. State whether your answer agrees with the power rule.

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$$

Answer: ___________________

4. Find y' using the product rule. $y = (x^2 + 3)(2x - 5)$

Answer: ___________________

5. Find $f'(x)$ using the quotient rule, and simplify the numerator completely. $f(x) = \dfrac{3x-1}{x+4}$

Answer: __________________

6. Find $h'(x)$. $h(x) = x^2 \cos x$

Answer: __________________

7. Find $f'(x)$ and simplify. $f(x) = \dfrac{x^2+1}{x^2-1}$

Answer: __________________

8. Let $f(x) = (2x+1)\sqrt{x}$.

(a) Find $f'(x)$ using the product rule, and write it with a single radical in each term.

(b) Find the slope of the tangent line to $y = f(x)$ at $x = 4$.

Answer: ________________

9. Let $f(x) = \dfrac{x^2}{2x-3}$.

(a) Find $f'(x)$ using the quotient rule. Leave the denominator squared.

(b) Evaluate $f'(1)$.

Answer: ________________

10. Find $f'(x)$. You will need the quotient rule on the outside and the product rule inside the numerator; you do not have to expand.

$$f(x) = \frac{x \sin x}{x + 1}$$

Answer: ____________________